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Exam 1

① HTGR's are gas cooled - helium cooled in modern designs - versus water cooled in LWRs. HTGR's operate at much higher temperatures, and therefore, can also be used for process heat, such as for endothermic reactions. HGTR benefits include fuel flexibility (most commonly UO_2 and UC fuels) and inherent safety due to the fuel particle design. Additionally, HGTRs can operate in a thermal or fast neutron spectrum and extensively utilize graphite.

② fuel kernel → where fission / heat generation occurs

buffer layer → porous graphite layer that provides space for kernel to swell and for fission product gas release (or CO production)

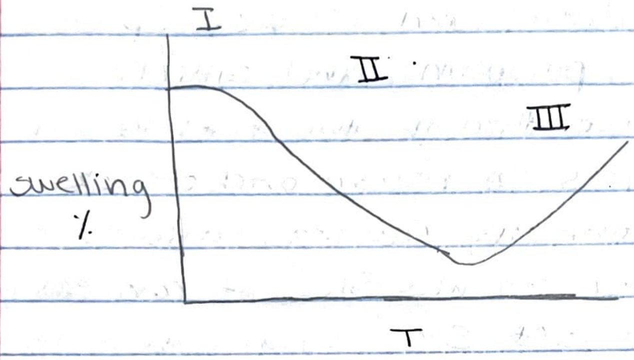
IPyC → protects SiC from fission product attack, keeps SiC in compressive state

SiC → primary pressure boundary / retains fission products

OPyC → protects SiC during manufacturing, keeps SiC in compressive state

③ Oxide fuel kernels are easier to manufacture and better at retaining fission products. However, fission generates free oxygen, which reacts with graphite to form CO gas, pressurizing the particle. Carbide fuel kernels demonstrate great high temperature stability and do not have the problem of CO generation, but are much worse at retaining fission products in the kernel.

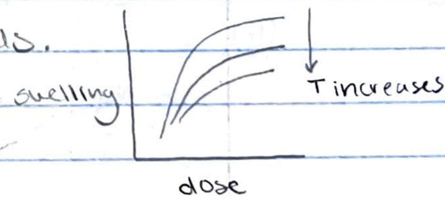
(4)



I = amorphous
II = point defect regime
III = void swelling regime

I: In this region, below about 423 K, the crystal will become amorphous with sufficient dose and high levels of swelling are observed.

II: In this region, between about 423K to 1273K, black spot defects dominate. As temperature increases, more recombination occurs and the swelling saturates at lower levels.



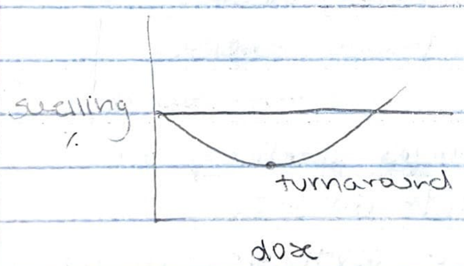
III: In this region, Frank loops and voids begin to form. Voids do not saturate, and the swelling continues to increase with increasing temperature.

(5) Thermal conductivity decreases with irradiation in SiC. Phonon-defect scattering is the primary mechanism. Point-defects are the largest source of phonon scattering, but this effect saturates as defect concentration increases. Voids also scatter phonons, although to a lesser degree, and do not saturate.

⑥ Fission products include soluble and insoluble oxides, noble gasses, metals, and volatiles. In an oxide fuel kernel, many fission products are trapped as oxides. Insoluble FPs can migrate to grain boundaries and escape. Specific FPs of concern are paladium and silver. Paladium can diffusive through the fuel kernel and the graphite layers to reach and attack the SiC layer, potentially causing failure. Additionally, if silver reaches SiC, it can easily diffuse through an intact SiC layer, allowing this FP to escape the particle boundary.

Noble gas fission products typically remain within the fuel kernel, forming voids / bubbles intra- or inter- granually.

⑦



Graphite is a porous material due to unfilled voids, trapped gasses, and thermal cracks during manufacturing. Thermal cracks arise due to the anisotropic thermal expansion coefficient during the cooling process, which results in small cracks parallel to the basal planes. During irradiation, interstitial defects form and preferentially deposit between the basal planes. These defects induce an increase in c, but a decrease in a. As a result of the thermal cracks between basal planes, the c direction can accommodate this swelling, and the shrinkage in a produces a net volume decrease.

Once the porosity has been filled, the defect-produced increase in the c parameter will cause swelling in the net volume. This point is called graphite turnaround, when the graphite goes from shrinking to swelling.

8) 1. overpressurization → Excess gaseous fission products, or CO production in oxide fuels, can result in too high of an internal pressure and result in failure of ~~the~~ SiC layer.

2. fission product attack of SiC → If FPs are able to reach the SiC layer, they can eat away and thin out the SiC and eventually cause failure.

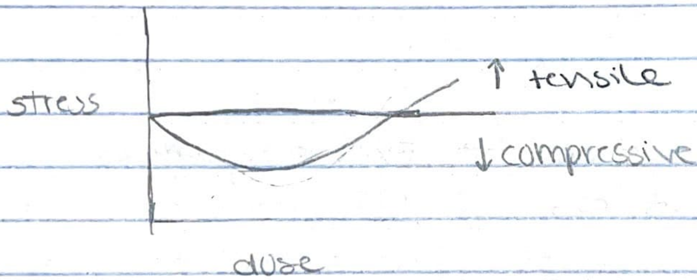
3. Kernel migration → A temperature gradient in oxide fuels can cause migration of the kernel to the lower temperature side, imparting high stresses on the layers.

(9) An advanced TRISO concept is a ZrC layer instead of SiC or in addition to SiC.

ZrC is stable at higher temperatures than SiC.

Another advanced concept is the addition of more graphite layers between the fuel kernel and SiC layer to better protect it from FP attack.

(10)



Initially, the SiC is in a compressive state due to the forces exerted by the IPyC and OPyC shrinkage. However, as gaseous fission products (or CO) build up and increase the internal pressure, a larger tensile force is exerted. Eventually, with this internal pressure and potential graphite turnaround, the tensile forces dominate.